

LECTURE 07

CONDITIONAL FORMATTING IN EXCEL

Highlight What Matters.
Analyze Smarter. Decide Faster.



TURN RAW DATA
INTO POWERFUL
BUSINESS INSIGHTS

Order ID	Region	Salesperson	Product Category	Revenue	Profit %	Status	Rating
ORD-1001	North	Aisha Khan	Electronics	125,000	↑ 28%	Completed	4.6
ORD-1002	South	Rohan Verma	Furniture	48,500	↑ 12%	Completed	★ ★ ★ ★
ORD-1003	East	Priya Singh	Clothing	215,000	↓ 32%	Completed	★ ★ ★ ★
ORD-1004	West	Arun Mehta	Electronics	85,200	↓ 18%	Pending	★ ★ ★
ORD-1005	North	Neha Sharma	Furniture	162,300	↓ 25%	Completed	★ ★ ★
ORD-1006	South	Karan Patel	Clothing	22,450	↓ 8%	Cancelled	★ ★
ORD-1007	East	Simran Kaur	Electronics	195,600	↑ 30%	Completed	★ ★ ★
ORD-1008	West	Vivek Joshi	Furniture	67,800	↓ 15%	Pending	★ ★ ★
ORD-1009	North	Aisha Khan	Clothing	112,400	↓ 22%	Completed	★ ★ ★
ORD-1010	South	Rohan Verma	Electronics	31,500	↓ 10%	Cancelled	★ ★

WHAT IS CONDITIONAL FORMATTING?

Conditional Formatting is an Excel feature that automatically changes the appearance of cells based on specific conditions or rules. It helps you quickly identify trends, patterns, and exceptions in your data.



WHY DO BUSINESSES USE IT?



QUICKLY
IDENTIFY
IMPORTANT
DATA



SAVE TIME
IN DATA
ANALYSIS



SPOT TRENDS,
PATTERNS &
EXCEPTIONS



ENHANCE
REPORTS &
DASHBOARDS



SUPPORT
FASTER
DECISIONS



IMPROVE DATA
QUALITY &
ACCURACY

REAL BUSINESS USE CASES



HIGH PERFORMERS
Highlight top sales,
profit, products,
or employees.



EXCEPTIONS
Spot low inventory,
negative profits,
or overdue orders.



DATA QUALITY
Find duplicate entries,
invalid data, or
missing values.



CUSTOMER INSIGHTS
Identify high-value
customers and
low satisfaction.



GOAL TRACKING
Monitor targets,
achievement %, and
performance gaps.



TIME-BASED ALERTS
Highlight today's orders,
upcoming deadlines,
or expiring items.

“ Good Analysts find data. Great Analysts **highlight** what matters. ”



Lecture 07 – Conditional Formatting in Excel

Module Objective

- Understand Conditional Formatting and its importance.
- Highlight important business insights automatically.
- Identify trends, exceptions, and outliers.
- Build visually appealing reports and dashboards.
- Apply industry-standard formatting rules used by Data Analysts.

Module 1 – Introduction to Conditional Formatting

What is Conditional Formatting?

Conditional Formatting is an Excel feature that automatically changes the appearance of cells when they satisfy specific conditions or rules.

Instead of manually coloring cells, Excel evaluates the data continuously and applies formatting dynamically whenever the specified condition is met.

Why Do Businesses Use Conditional Formatting?

In real-world datasets, companies may have thousands of transactions. Reading each record manually is inefficient.

Conditional Formatting helps analysts quickly identify:

- High-performing products
- Poor-performing regions
- Delayed orders
- Duplicate entries
- Top customers
- Low inventory
- High-risk transactions

without writing complex formulas.

Business Example

Suppose a company has **5,000 sales records**.

The Sales Manager asks:

"Show me only orders where Revenue is greater than ₹100,000."

Instead of filtering repeatedly, Conditional Formatting instantly highlights those rows.

Result:

The manager can immediately identify high-value orders.

Benefits of Conditional Formatting

- Improves report readability.
- Saves analysis time.
- Highlights important business insights.
- Makes dashboards interactive.
- Helps management make faster decisions.



- Reduces manual checking.

Module 2 – Types of Conditional Formatting

Excel provides several built-in Conditional Formatting categories:

1. Highlight Cells Rules
2. Top/Bottom Rules
3. Data Bars
4. Color Scales
5. Icon Sets
6. New Rule (Formula-Based Formatting)
7. Manage Rules
8. Clear Rules

We will study each one in detail.

Module 3 – Highlight Cells Rules

These rules highlight cells based on their values.

Used extensively in business reporting.

1. Greater Than

Highlights cells whose values are greater than a specified number.

Business Scenario

The Sales Director wants to identify orders with Revenue greater than ₹100,000.

Revenue Column

45000

92000

135000

81000

155000

Condition

Greater Than 100000

Highlighted

135000

155000

Why is this useful?

Instead of filtering the report every day, management immediately sees premium sales.

Business Use Cases

- High Revenue Orders
- High Profit Products
- Top Salespersons



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- High Customer Ratings

2. Less Than

Definition

Highlights values smaller than a given number.

Business Scenario

Finance wants to identify products generating revenue below ₹20,000.

These products may require promotions.

Business Use Cases

- Low Revenue
- Low Stock
- Low Ratings
- Low Attendance

3. Between

Highlights values falling within a specified range.

Business Example

Highlight Profit %
between
20%
and
30%

Why?

Management wants to identify medium-performing products.

Business Uses

- Salary ranges
- Revenue bands
- Discount ranges
- Performance categories

4. Equal To

Highlights values exactly matching a specific value.

Business Scenario

Highlight all orders where
Status = Pending

Business Uses

- Pending Orders



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- Cancelled Orders
- Returned Products
- VIP Customers

5. Text that Contains

Highlights text containing a specified word.

Business Example

Highlight every product containing
Laptop

Business Uses

- Product Search
- Employee Name Search
- City Search
- Department Identification

6. A Date Occurring

Definition

Highlights dates matching predefined periods.

Examples

- Today
- Yesterday
- Last Week
- Next Month
- This Month

Business Example

Highlight all orders placed today.

Business Uses

- Today's Orders
- Expiring Contracts
- Upcoming Meetings
- Invoice Due Dates

7. Duplicate Values

Definition

Highlights repeated values.

Business Example

Customer ID
C101
C105

C109
C105
Duplicate
C105

Why?

Duplicate IDs often indicate data quality issues.

Business Uses

- Duplicate Customer IDs
- Duplicate Invoice Numbers
- Duplicate Employee IDs
- Duplicate Product Codes

Module 4 – Top/Bottom Rules

These rules identify the highest and lowest performing records.

1. Top 10 Items

Highlights the highest values.

Business Example

Top 10 Revenue transactions.

Business Uses

- Best Customers
- Best Products
- Highest Revenue Orders
- Top Salespersons

2. Bottom 10 Items

Highlights the lowest values.

Business Example

Lowest Revenue Products.

Business Uses

- Poor Sales
- Weak Products
- Low Performance Regions

3. Top 10%

Highlights the highest percentage of data.



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Business Example
Top 10% customers by Revenue.

4. Bottom 10%

Highlights the lowest percentage.
Useful for identifying underperforming records.

5. Above Average

Highlights values greater than the dataset average.

Business Example

Revenue
Average
₹68,000
Highlight
All values above ₹68,000.

Business Uses

Sales Analysis
Profit Analysis
Customer Spending

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6. Below Average

Highlights values below the average.
Useful for identifying weak performers.

Module 5 – Data Bars

Data Bars create horizontal bars inside cells proportional to their values.
Larger values receive longer bars.

Business Example

Revenue
25000
52000
89000
145000
Excel automatically creates bars of varying lengths.

Why?

Humans compare lengths much faster than numbers.



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Business Uses

- Revenue Comparison
- Sales Quantity
- Inventory
- Profit

Module 6 – Color Scales

Applies gradient colors based on values.

Typically:

Green → High

Yellow → Medium

Red → Low

Business Example

Customer Ratings

2.5

3.8

4.9

Higher ratings appear greener.

Business Uses

- KPI Dashboards
- Customer Satisfaction
- Performance Reports

Module 7 – Icon Sets

Uses icons instead of colors.

Examples



Traffic Lights

Arrows

Flags

Business Example

Profit %

Green Arrow

High Profit

Yellow Arrow



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Average
Red Arrow
Low Profit

Business Uses

Executive Dashboards
Performance Reports
Financial KPIs
Target Achievement

Module 8 – Formula-Based Conditional Formatting

Why Formula-Based Rules?

Built-in rules are limited.

Business scenarios often require custom logic.

Example 1

Highlight entire row when

Status

=

Cancelled

Formula

= \$K2="Cancelled"

Example 2

Highlight Revenue greater than ₹100,000

AND

Profit > 20%

Formula

=AND(\$I2>100000,\$J2>20)

Example 3

Weekend Orders

=WEEKDAY(\$B2,2)>5

Business Uses

- High-value profitable orders
- Pending high-priority transactions
- Weekend sales
- Regional analysis
- Dynamic dashboards



Module 9 – Managing Rules

After creating multiple rules, analysts must manage them properly.

Topics Covered

- Edit Rule
- Delete Rule
- Rule Priority
- Stop If True
- Applies To Range

Why is Rule Management Important?

Large enterprise dashboards may contain **50–100 Conditional Formatting rules**. Poor rule management can lead to conflicting colors, incorrect highlights, and slower workbook performance.

Understanding rule priority ensures that the correct formatting is applied consistently across reports.

Best Practices

- Use Conditional Formatting to highlight insights, not decorate worksheets.
- Avoid applying too many colors simultaneously.
- Use consistent color meanings (e.g., green = good, red = bad).
- Prefer formula-based rules for complex business logic.
- Apply formatting only to the required range to maintain workbook performance.
- Combine Conditional Formatting with Tables, Pivot Tables, and Dashboards for interactive reporting.



Part 02 – Advanced Conditional Formatting & Professional Business Applications

Module 10 – New Rule

What is "New Rule"?

The **New Rule** option allows users to create custom Conditional Formatting rules that are **not available** in Excel's built-in formatting options.

Unlike Highlight Cell Rules or Top/Bottom Rules, **New Rule** provides complete flexibility by allowing formulas, custom conditions, and formatting logic.

Why Do Data Analysts Use New Rules?

Built-in formatting works for simple conditions.

Business requirements are often much more complex.

For example,

Highlight orders where Revenue > ₹100,000 **AND** Profit > 25%.

This cannot always be achieved efficiently using basic rules.

New Rule solves this problem.

Types of New Rules

Excel provides several rule types.

1. Format All Cells Based on Their Values

Purpose

Applies formatting according to the cell values.

Usually used with

- Data Bars
- Color Scales
- Icon Sets

Business Example

Revenue Column

₹12,000

₹38,000

₹65,000

₹1,10,000

₹2,45,000

Higher Revenue

↓
Darker Green
Lower Revenue
↓
Lighter Green

Business Use Cases

- Revenue
- Profit
- Inventory
- Customer Spending
- Sales Quantity

2. Format Only Cells That Contain

Purpose

Formats only those cells satisfying a specified condition.

Example

Highlight

Revenue

Greater Than

₹100,000

Business Scenario

Management wants to identify premium orders.

Practical Uses

- High Revenue
- High Discounts
- Low Inventory
- Low Ratings

3. Format Only Top or Bottom Ranked Values

Purpose

Highlights top or bottom performers.

Example

Top 5 Revenue Orders

Bottom 10 Profit %

Business Uses

- Best Salespersons
- Best Products
- Weak Products



- Low Performing Regions

4. Format Only Values Above or Below Average

Purpose

Highlights values relative to the average.

Business Example

Average Revenue

₹68,000

Highlight every revenue above average.

Why?

Managers instantly identify better-performing transactions.

5. Format Only Unique or Duplicate Values

Purpose

Detect duplicate records.

Business Example

Duplicate Invoice Number

Duplicate Customer ID

Duplicate Employee ID

Business Importance

Duplicate records create reporting errors.

Analysts detect them immediately.

6. Use a Formula to Determine Which Cells to Format

This is the most important option.

Nearly every Data Analyst uses Formula-Based Conditional Formatting.

Module 11 – Formula-Based Conditional Formatting

Excel evaluates a logical formula.

If the formula returns

TRUE

↓

Formatting is applied.

If

FALSE

↓

Nothing happens.



Formula Rules Always Return

TRUE

or

FALSE

Business Logic

Suppose

Revenue

Column

I

Profit %

Column

J

Requirement

Highlight orders where

Revenue > ₹100,000

AND

Profit > 20%

Formula

=AND(\$I2>100000,\$J2>20)

Explanation

Revenue greater than

100000

AND

Profit above

20%

↓

Highlight row.

Business Benefit

Identify

High Revenue

•

High Profit

Orders

Instantly.

Formula Example 2

Highlight Cancelled Orders

Formula

= \$K2="Cancelled"

Explanation



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Status column

contains

Cancelled



Entire row becomes Red.

Business Uses

Cancelled Orders

Pending Orders

Delayed Deliveries

Rejected Claims

Formula Example 3

Weekend Orders

Formula

`=WEEKDAY($B2,2)>5`

Explanation

Saturday

Sunday



Highlighted

Business Uses

Weekend Sales

Weekend Attendance

Weekend Support Tickets

Formula Example 4

Low Customer Rating

Formula

`=L2<3`

Business Use

Immediately identify dissatisfied customers.

Formula Example 5

Top Salesperson

Formula

`=I2=MAX($I$2:$I$101)`

Explanation

Highest Revenue



Automatically highlighted.



Formula Example 6

Today's Orders

Formula

=B2=Today()

Business Uses

Today's Deliveries

Today's Orders

Today's Complaints

Today's Appointments

Formula Example 7

Upcoming Deliveries

Formula

=AND(B2>=Today(),B2<=Today()+7)

Business Use

Highlight deliveries within

7 days.

Formula Example 8

Expired Contracts

Formula

=B2<Today()

Business Uses

Expired Licenses

Expired Contracts

Overdue Invoices

Subscription Renewal

Module 12 – Dynamic KPI Highlighting

What is KPI?

KPI

=

Key Performance Indicator

Businesses monitor

Revenue

Profit

Growth

Orders

Customers

Targets



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using KPIs.

Conditional Formatting makes KPIs interactive.

Business Example

Revenue

Target

₹100,000

Above Target

↓

Green

Below Target

↓

Red

Manager instantly understands performance.

KPI Example

Profit %

Greater than
25%

↓

Green

Between
15–25%

↓

Yellow

Below

15%

↓

Red

Reason

Traffic-light indicators improve decision-making.

Module 13 – Heat Maps

Heat Maps use Color Scales to visualize data density.

Higher values

↓

Dark Green

Lower values

↓



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Light Red

Business Example

Monthly Sales

Jan Feb Mar Apr

45K 62K 95K 1.2L

Immediately shows best-performing months.

Business Uses

Sales Trends

Customer Spending

Inventory

Website Traffic

Revenue Analysis

Module 14 – Progress Tracking

Conditional Formatting can create progress bars.

Business Example

Target

₹500,000

Achieved

₹420,000

Progress Bar



84%

Uses

Sales Targets

Project Completion

Employee Goals

Budget Utilization

Module 15 – Dashboard Applications

Professional dashboards heavily depend on Conditional Formatting.

Example Dashboard

Revenue KPI

● Above Target

Profit KPI

● Medium

Customer Rating



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● Low
Inventory
△ Low Stock
Orders
↑ Increasing

Managers can understand the entire business within seconds.

Module 16 – Conditional Formatting Best Practices

Do

- ✓ Keep color meanings consistent.
- ✓ Use Green for good.
- ✓ Use Red for issues.
- ✓ Use Yellow for warnings.
- ✓ Use Formula-Based Rules whenever business logic becomes complex.
- ✓ Test every rule.
- ✓ Keep dashboards clean.

Don't

- ✗ Use too many colors.
- ✗ Highlight every column.
- ✗ Apply rules to unnecessary ranges.
- ✗ Mix random formatting styles.
- ✗ Create overlapping conflicting rules.

Real Business Case Study

Scenario

An e-commerce company receives **15,000 orders per month**.

The management wants the report to automatically:

- Highlight revenue above ₹100,000.
- Show cancelled orders in red.
- Mark pending orders in yellow.
- Flag customer ratings below 3.
- Highlight duplicate Order IDs.
- Show top-performing salespersons.
- Display progress toward monthly revenue targets.

Using Conditional Formatting, the analyst creates an interactive report where important records stand out instantly without manually reviewing thousands of rows.



Interview-Oriented Questions (with Answers)

Q1. What is Conditional Formatting?

Answer: A feature in Excel that automatically changes cell formatting based on specified conditions or formulas, helping users identify trends, patterns, and exceptions.

Q2. Why is Conditional Formatting important for Data Analysts?

Answer: It improves data visualization, highlights critical insights, identifies anomalies, supports dashboards, and speeds up business decision-making.

Q3. What is the difference between Highlight Cell Rules and Formula-Based Conditional Formatting?

Answer: Highlight Cell Rules apply predefined conditions (e.g., Greater Than, Duplicate Values), while Formula-Based Conditional Formatting allows custom business logic using formulas that return TRUE or FALSE.

Q4. Why are absolute references (\$) commonly used in Conditional Formatting formulas?

Answer: Absolute references lock specific columns or rows so the rule evaluates the correct field while being applied across multiple cells or rows.

Q5. How would you highlight an entire row when the Status is "Cancelled"?

Answer:

`=K2="Cancelled"`

Q6. Which function can highlight only weekend dates?

Answer:

`=WEEKDAY(A2,2)>5`

Q7. How can Conditional Formatting improve dashboards?

Answer: By visually emphasizing KPIs, targets, trends, exceptions, and performance using colors, icons, and data bars, making dashboards easier to interpret.

Q8. What are Heat Maps in Excel?

Answer: Heat Maps use Color Scales to represent data intensity, allowing users to identify high and low values at a glance.

Q9. What is the purpose of the "Manage Rules" option?

Answer: It allows users to edit, delete, reorder, and control the priority of Conditional Formatting rules to prevent conflicts.

Q10. Give a real-world business example of Formula-Based Conditional Formatting.

Answer: Highlight all orders where **Revenue > ₹100,000** and **Profit > 20%** to identify high-value, high-profit transactions requiring management attention.

